3. Random Vectors and Joint Distributions

3.1 Definitions and Multivariate CDFs

- In random experiments there usually are multiple random variables of interest associated with each possible outcome ω. Here are some examples.
  i. A political pollster will usually ask several questions: who you are likely to vote for, which party do you prefer, are you male or female, etc.
  ii. The response of a patient to a drug typically depends on other variables besides disease status, such as age, physical condition, gender, and so on.
  iii. For each node of a computer network we can observe various quantities such as the number of connections and duration of the connection(s).
- For statistics, we not only have multiple variables being observed, we also have multiple individuals in the study. Even with just one measure per individual, the experiment yields a vector of values.
- Moreover, statistical analysis usually results in multiple statistics such as sample means, sample standard deviations and other estimates.

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- A vector $(X_1, \ldots, X_k)$ of random variables defined on the same sample space Ω is called a *random vector*.
- The *probability distribution* of $(X_1, \ldots, X_k)$ is a rule that defines $\mathsf{P}((X_1, \ldots, X_k) \in B)$ for all $B \subset \mathbb{R}^k$. Such distributions are often called *multivariate* or *joint*.
- A joint distribution is always determined by its *joint cumulative distribution function* (cdf)

$$F(x_1, \ldots, x_k) = \mathsf{P}(X_1 \leq x_1, \ldots, X_k \leq x_k) \quad \text{for } x_i \in \mathbb{R},\ i = 1, \ldots, k.$$

- In the context of a multivariate distribution, the distribution of each X_i is called its *marginal distribution*.
- The distribution of any *sub-vector* $(X_{i_1}, \ldots, X_{i_m})$ is sometimes also called a marginal distribution.

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- The joint distribution of $(X_1, \ldots, X_k)$ is called

  i. *discrete* if there is a *countable* set $R \subset \mathbb{R}^k$ such that

  $$\mathsf{P}((X_1, \ldots, X_k) \in R) = 1,$$

  ii. *continuous* if the joint cdf is a *continuous* function in $\mathbb{R}^k$,

  iii. *absolutely continuous* if there exists a function $f(x_1, \ldots, x_k)$ such that

  $$\mathsf{P}((X_1, \ldots, X_k) \in A) = \int_A \cdots \int f(x_1, \ldots, x_k)\, \mathrm{d}x_1 \cdots \mathrm{d}x_k \quad \text{for all } A.$$

  iv. There are other types of distributions, including *mixed* and *singular continuous*. In fact, complicated distributions can come about in the multivariate context even more easily than they do in the univariate context.

- As before we will focus on discrete and absolutely continuous cases.

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Bivariate Distributions

For simplicity, suppose there are only two random variables, say X and Y, of interest and let

$$F_{X,Y}(x,y) = \mathsf{P}(X \le x, Y \le y) = \mathsf{P}(\{X \le x\} \cap \{Y \le y\})$$

be their joint cumulative distribution function.

As always, the distinction between strict ($<$) and non-strict ($\le$) inequalities can matter.

Obtain the probability that (X,Y) falls in a rectangle

$$(a,b] \times (c,d] = \{(x,y) : a < x \le b, c < y \le d\}$$

using just the cdf.

Solution: Start by drawing a picture of the region under consideration.

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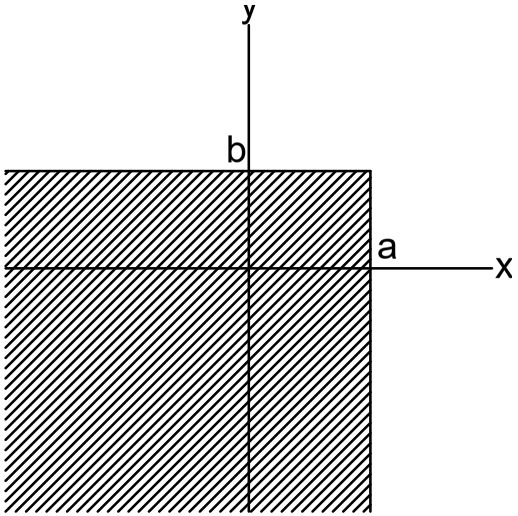
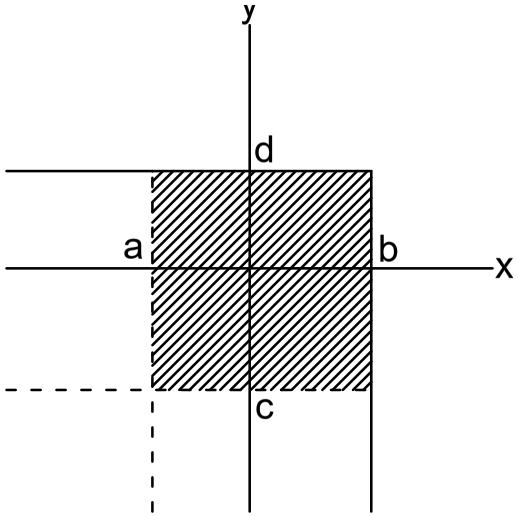
Calculating a Probability from a Bivariate CDF

F(a,b)=prob of shaded area



P(a<X<=b,c<Y<=d)= prob of shaded area



$$
\begin{aligned}
&\mathsf{P}(a < X \le b, c < Y \le d) \\
&\quad = \mathsf{P}(X \le b, Y \le d) - \mathsf{P}(X \le a, Y \le d) - \mathsf{P}(a < X \le b, Y \le c) \\
&\quad = F(b,d) - F(a,d) - (\mathsf{P}(X \le b, Y \le c) - \mathsf{P}(X \le a, Y \le c)) \\
&\quad = F(b,d) - F(a,d) - F(b,c) + F(a,c).
\end{aligned}
$$

Compare this to the univariate result: $\mathsf{P}(a < X \le b) = F_X(b) - F_X(a)$.

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Bivariate Distribution Functions

- For each fixed y, $F_{X,Y}(x,y) = \mathsf{P}(X \le x, Y \le y)$ is a *nondecreasing* function of x. (Recall the corresponding property for the cdf of a single random variable.)

  Likewise, $F_{X,Y}(x,y)$ is a nondecreasing function of y for each fixed x.

- However, those conditions are not sufficient. In view of the computation on the previous slide, it is also necessary that

  $$F_{X,Y}(b,d) - F_{X,Y}(a,d) - F_{X,Y}(b,c) + F_{X,Y}(a,c) \ge 0 \quad \text{for all } a < b,\ c < d.$$

  For example, $F(x,y) = x + y - xy$, with $0 \le x \le 1$ and $0 \le y \le 1$, is *not a bivariate cdf* even though it is nondecreasing in each variable and $0 \le F(x,y) \le 1$.

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- Some limit properties of a bivariate cdf.

  i. The marginal cdf for X can be found by

$$F_X(x) = \mathsf{P}(X \leq x, Y < \infty) = \lim_{y\to\infty} F_{X,Y}(x,y).$$

  Likewise, $F_Y(y) = \lim_{x\to\infty} F_{X,Y}(x,y)$. Consequently,

$$\lim_{x\to\infty}\lim_{y\to\infty} F_{X,Y}(x,y) = 1.$$

  ii. On the other hand,

$$\lim_{x\to-\infty} F(x,y) = \lim_{y\to-\infty} F_{X,Y}(x,y) = 0.$$

  iii. Continuity "from above":

$$\lim_{u\downarrow x}\lim_{v\downarrow y} F_{X,Y}(u,v) = F_{X,Y}(x,y).$$

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3.2 Discrete Bivariate Distributions

- (X, Y) is discrete if takes a countable number of values (in $\mathbb{R}^2$), with probability 1. Clearly, this implies X and Y are each discrete in the sense of the previous chapter. So it suffices to consider all points (x, y) where $x \in R_X$ and $y \in R_Y$ (the respective ranges of X and Y).
- In fact, the distribution is completely described by knowledge of its *joint probability mass function*

$$p_{X,Y}(x, y) = \mathsf{P}(\{X = x\} \cap \{Y = y\}) = \mathsf{P}(X = x, Y = y)$$

  for all ordered pairs (x, y).
- Since, in the discrete case, there are only countably many ordered pairs such that $p_{X,Y}(x, y) > 0$, the sum over all such pairs must equal 1:

$$\sum\sum_{(x,y)} p_{X,Y}(x, y) = 1.$$

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Discrete Bivariate Distribution (cont.)

- And, for any two-dimensional set A, we have

$$\mathsf{P}((X,Y) \in A) = \sum\sum_{(x,y)\in A} p_{X,Y}(x,y).$$

Order of summation does not matter since p is nonnegative.

- In particular, the joint cdf is

$$F_{X,Y}(x,y) = \sum_{u:u\le x}\sum_{v:v\le y} p_{X,Y}(u,v) \quad \text{for all real } x, y.$$

Note the care taken to avoid using the same letter multiple ways.

- If $g(x,y)$ is a nonnegative function such that $C = \sum\sum_{(x,y)} g(x,y) < \infty$ then $p(x,y) = \frac{g(x,y)}{C}$ is a joint pmf for some random pair (X,Y).

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Discrete Bivariate Distribution (cont.)

- The *marginal* distributions can be found from the joint pmf. For example,

$$p_X(x) = \mathsf{P}(X = x, Y < \infty) = \sum_y p_{X,Y}(x, y)$$

and

$$\begin{aligned} F_X(x) &= \sum_{u:u \le x} p_X(u) = \sum_{u:u \le x} \sum_y p_{X,Y}(u, y) \\ &= \sum_y \sum_{u:u \le x} p_{X,Y}(u, y). \end{aligned}$$

Likewise, $p_Y(y) = \sum_x p_{X,Y}(x, y)$ and

$$F_Y(y) = \sum_{v:v \le y} p_Y(v) = \sum_x \sum_{v:v \le y} p_{X,Y}(x, v).$$

- However, $p_{X,Y}(x, y)$ cannot be computed just from $p_X(x)$ and $p_Y(y)$, in general. (More on this later – meanwhile consider the fact that $p_{X,Y}(x, y)$ is the probability of the intersection of events $\{X = x\}$ and $\{Y = y\}$, which need not be independent.)

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Hearts

Example 3–1. Consider a sequential experiment consisting of rolling a fair die and observing value X, and then drawing cards $\lfloor \frac{X+1}{2} \rfloor$ cards randomly, *with replacement*, from a standard deck. That is, if $X = 1$ or 2 then we draw one card, if $X = 3$ or 4 then we draw two; and otherwise we draw three. Let Y be the number of "Hearts" cards selected.

Find the joint probability mass function for (X, Y).

Solution: Clearly, if k cards are drawn then the number of Hearts is a binomial$(k, \frac{1}{4})$ random variable. So we can use conditional probability to determine that if $x =$ 1 or 2 then

$$p_{X,Y}(x, y) = \mathsf{P}(X = x)\mathsf{P}(Y = y|X = x) = \tfrac{1}{6}(\tfrac{1}{4})^y(\tfrac{3}{4})^{1-y} \quad \text{for } y = 0, 1.$$

Likewise, if $x = 3$ or 4 then $p_{X,Y}(x, y) = \frac{1}{6}\binom{2}{y}(\frac{1}{4})^y(\frac{3}{4})^{2-y}$ for $y = 0, 1, 2$.

And if $x = 5$ or 6 then $p_{X,Y}(x, y) = \frac{1}{6}\binom{3}{y}(\frac{1}{4})^y(\frac{3}{4})^{3-y}$ for $y = 0, 1, 2, 3$.

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Hearts (cont.)

Computing, we get the following table of the probability mass function for (X, Y):

		y			
		0	1	2	3
	1	$\frac{1}{8}$	$\frac{1}{24}$	0	0
	2	$\frac{1}{8}$	$\frac{1}{24}$	0	0
x	3	$\frac{3}{32}$	$\frac{1}{16}$	$\frac{1}{96}$	0
	4	$\frac{3}{32}$	$\frac{1}{16}$	$\frac{1}{96}$	0
	5	$\frac{9}{128}$	$\frac{9}{128}$	$\frac{9}{128}$	$\frac{1}{384}$
	6	$\frac{9}{128}$	$\frac{9}{128}$	$\frac{9}{128}$	$\frac{1}{384}$

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Hearts (cont.)

Find the marginal distributions of X and Y.

Solution: Sum the rows and columns.

		y				
		0	1	2	3	$p_X(x)$
x	1	$\frac{1}{8}$	$\frac{1}{24}$	0	0	$\frac{1}{6}$
	2	$\frac{1}{8}$	$\frac{1}{24}$	0	0	$\frac{1}{6}$
	3	$\frac{3}{32}$	$\frac{1}{16}$	$\frac{1}{96}$	0	$\frac{1}{6}$
	4	$\frac{3}{32}$	$\frac{1}{16}$	$\frac{1}{96}$	0	$\frac{1}{6}$
	5	$\frac{9}{128}$	$\frac{9}{128}$	$\frac{3}{128}$	$\frac{1}{384}$	$\frac{1}{6}$
	6	$\frac{9}{128}$	$\frac{9}{128}$	$\frac{3}{128}$	$\frac{1}{384}$	$\frac{1}{6}$
	$p_Y(y)$	$\frac{111}{192}$	$\frac{67}{192}$	$\frac{13}{192}$	$\frac{1}{192}$	1

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Example 3–2. A poker hand of 5 cards is selected at random from an ordinary deck of 52 cards. Let X = number of aces and Y = number of face cards in the hand. Find the joint pmf for (X, Y).

Solution: First, *determine the possible values for X and Y*. X must be less than or equal to 4, while Y can be any integer from 0 to 5. But, also, their sum can be no more than 5.

We recall that there are $\binom{52}{5}$ possible poker hands. The number of ways to select x aces, out of the four in the deck, is $\binom{4}{x}$ and the number of ways to select y face cards is $\binom{12}{y}$. The other $5 - x - y$ cards come from the remaining 36 in the deck, giving us the factor $\binom{36}{5-x-y}$.

Hence, the probability we want is

$$p_{X,Y}(x, y) = \mathsf{P}(X = x, Y = y) = \frac{\binom{4}{x}\binom{12}{y}\binom{36}{5-x-y}}{\binom{52}{5}},$$

for $0 \le x \le 4$, $0 \le y \le 5 - x$, x and y both integers.

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Aces and Faces (cont.)

What is the chance the hand has a total of 3 aces and face cards?

Solution:

$$\begin{aligned}
\mathsf{P}(X+Y=3) &= p_{X,Y}(3,0)+p_{X,Y}(2,1)+p_{X,Y}(1,2)+p_{X,Y}(0,3) \\
&= \frac{\binom{4}{3}\binom{12}{0}\binom{36}{2}}{\binom{52}{5}} + \frac{\binom{4}{2}\binom{12}{1}\binom{36}{2}}{\binom{52}{5}} + \frac{\binom{4}{1}\binom{12}{2}\binom{36}{2}}{\binom{52}{5}} + \frac{\binom{4}{0}\binom{12}{3}\binom{36}{2}}{\binom{52}{5}} \\
&= 0.13575.
\end{aligned}$$

What is the chance the hand has more aces than face cards?

Solution:

$$\mathsf{P}(X>Y) = \sum_{(x,y):y<x} p_{X,Y}(x,y) = \sum_{x=1}^{4}\sum_{y=0}^{x-1} p_{X,Y}(x,y) = \cdots = 0.12635.$$

Check: Compute the sum in the other order: $\sum_{y=0}^{2}\sum_{x=y+1}^{4} p_{X,Y}(x,y)$.

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Aces and Faces (cont.)

Without considering Y, since X is the number of aces we must have

$$p_X(x) = \mathsf{P}(X = x) = \frac{\binom{4}{x}\binom{48}{5-x}}{\binom{52}{5}},$$

for integer x, $0 \leq x \leq 4$. This is a hypergeometric$(5, 52, 4)$ pmf. It is therefore the marginal pmf for X.

Likewise, $Y \sim$ hypergeometric$(5, 52, 12)$.

What is the distribution of $Z = 5 - X - Y$, which is the number of cards that are neither an ace or a face?

This is an example of sampling without replacement and observing the counts for multiple disjoint categories. It extends the hypergeometric model for which there are only two categories.

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1's and 6's

Example 3–3. Roll a fair die n times and observe $X =$ number of 1's and $Y =$ number of 6's. Find the joint pmf for (X, Y).

Solution: This is similar to working out the pmf of a binomial random variable. First, determine the possible values. Clearly, $0 \leq x \leq n$ and $0 \leq y \leq n$.

But we also must have $x + y \leq n$.

Second, the probability of a *particular* sequence of x 1's and y 6's (and *hence* $n - x - y$ *others*) is $(\frac{1}{6})^x(\frac{1}{6})^y(\frac{2}{3})^{n-x-y}$.

There are $\binom{n}{x}$ ways to arrange the x 1's. Given those choices, there are $\binom{n-x}{y}$ ways to arrange the y 6's. That means there are

$$\binom{n}{x}\binom{n-x}{y} = \frac{n!}{x!y!(n-x-y)!}$$

arrangements altogether, each with the probability above.

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1's and 6's (cont.)

Summarizing, we have shown

$$\mathsf{P}(X = x, Y = y) = \frac{n!}{x!y!(n-x-y)!}\left(\frac{1}{6}\right)^x\left(\frac{1}{6}\right)^y\left(\frac{2}{3}\right)^{n-x-y},$$

for $x \geq 0$, $y \geq 0$, $x + y \leq n$. This is an example of a *trinomial* distribution.

Here we have the joint distribution of counts from multiple categories when the selections are independent, extending the binomial case for two categories.

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1's and 6's (cont.)

Now find the marginal pmf for X.

Solution: In fact, ignoring Y, we see that X is a binomial$(n, \frac{1}{6})$ random variable.

Alternatively, we can sum the joint pmf over the possible values of y to get the marginal pmf for X.

Note that $X = x$ requires $0 \leq y \leq n - x$.

Suppose $0 \leq x \leq n$. Pulling out factors that depend only on x and then using *the binomial theorem* (with "n" replaced by "$n - x$"),

$$\begin{aligned} p_X(x) &= \sum_{y=0}^{n-x} p_{X,Y}(x,y) = \binom{n}{x}\left(\frac{1}{6}\right)^x \times \sum_{y=0}^{n-x} \binom{n-x}{y}\left(\frac{1}{6}\right)^y \left(\frac{2}{3}\right)^{n-x-y} \\ &= \binom{n}{x}\left(\frac{1}{6}\right)^x \left(\frac{1}{6} + \frac{2}{3}\right)^{n-x} \\ &= \binom{n}{x}\left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{n-x}, \end{aligned}$$

which is the binomial$(n, \frac{1}{6})$ pmf.

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3.3 Continuous Bivariate Distributions

- (X, Y) is absolutely continuous if there exists a function $f(x, y)$, called the *joint probability density function*, such that

$$\mathsf{P}((X, Y) \in A) = \iint_A f(x, y)\,\mathrm{d}x\mathrm{d}y$$

  for a subset A of the (x, y)-plane. So the probability that (X, Y) takes value in A is the *volume* under f that lies over that set.

  Order of integration does not matter since f is nonnegative.

- As in the univariate case, f must be nonnegative and integrate to 1:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\,\mathrm{d}x\mathrm{d}y = 1.$$

- If $g(x, y)$ is nonnegative and $C = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x, y)\,\mathrm{d}x\mathrm{d}y < \infty$ then $f(x, y) = \frac{g(x,y)}{C}$ is a pdf.

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Example 3–4. Consider the joint pdf for (X, Y):

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{x(1+3y^2)}{4} & 0 \le x \le 2, 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

We see that $f_{X,Y}(x,y) \ge 0$ for all (x,y) and that all the probability is on the *rectangle* $[0,2] \times [0,1]$.

Also,

$$\begin{aligned} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}x\mathrm{d}y &= \int_0^1\int_0^2 \frac{x(1+3y^2)}{4}\,\mathrm{d}x\mathrm{d}y \\ &= \int_0^1 \left(\left[\frac{x^2}{8} + \frac{3x^2y^2}{8}\right]\Big|_{x=0}^{x=2}\right)\mathrm{d}y = \int_0^1 \left(\frac{1}{2} + \frac{3y^2}{2}\right)\mathrm{d}y \\ &= \left[\frac{y}{2} + \frac{y^3}{2}\right]\Big|_{y=0}^{y=1} = \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

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Example 3–4 (cont.)

Find $\mathsf{P}(X > 1, Y \leq \frac{1}{2})$.

Solution: It usually helps to draw a picture of the region of integration. Keep in mind also, that it might be simpler to compute the complement probability.

You can integrate in either order (or *do both, as a check*).

$$\begin{aligned}
\mathsf{P}\Big(X > 1, Y \leq \frac{1}{2}\Big) &= \int_1^\infty \int_{-\infty}^{1/2} f_{X,Y}(x,y)\,\mathrm{d}y\mathrm{d}x \\
&= \int_1^2 \int_0^{1/2} \frac{x(1+3y^2)}{4}\,\mathrm{d}y\mathrm{d}x \\
&= \cdots = \int_1^2 \frac{x}{4}\left(\frac{1}{2}+\frac{1}{8}\right)\mathrm{d}x = \cdots = \frac{15}{64}\,.
\end{aligned}$$

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Example 3–4 (cont.)

Find $\mathsf{P}(X+Y \le 2)$.

Solution: Again, first draw a picture of the region of integration.

$$
\begin{aligned}
\mathsf{P}\Big(X+Y \le 2\Big) &= \int_0^2 \int_0^{2-x} f_{X,Y}(x,y)\,\mathrm{d}y\mathrm{d}x \\
&= \int_0^1 \frac{x}{4} \int_0^1 (1+3y^2)\,\mathrm{d}y\mathrm{d}x + \int_1^2 \frac{x}{4} \int_0^{2-x} (1+3y^2)\,\mathrm{d}y\mathrm{d}x \\
&= \cdots = \int_0^1 \frac{x}{2}\,\mathrm{d}x + \frac{1}{4}\int_1^2 x((2-x)+(2-x)^3)\,\mathrm{d}x \\
&= \frac{x^2}{4}\Big|_{x=0}^{x=1} + \frac{1}{4}\int_0^1 (2-v)(v+v^3)\,\mathrm{d}v = \cdots = \frac{59}{120}\,.
\end{aligned}
$$

Check: Integrate in the other order.

Check: Integrate to find $\mathsf{P}(X+Y>2)$ *instead, and subtract it from 1.*

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Bivariate Distribution Functions

- The joint *cumulative distribution function* (cdf) for an absolutely continuous random pair (X, Y) is

$$F_{X,Y}(x, y) = \mathsf{P}(X \leq x, Y \leq y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u, v)\, \mathrm{d}u\mathrm{d}v.$$

- Note that $\mathsf{P}(X \leq x, Y \leq y) = \mathsf{P}(X < x, Y < y)$, since the distribution is continuous.

- The pdf may be obtained from the cdf by differentiation:

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y}$$

  at all (x, y) where the *mixed partial derivative* exists. (Elsewhere, the pdf may be defined in any convenient way – it will not change the value of any integral.)

- As mentioned before, the probability that (X, Y) lies in a rectangle is given by

$$\mathsf{P}(a < X \leq b, c < Y \leq d) = F_{X,Y}(b, d) - F_{X,Y}(a, d) - F_{X,Y}(b, c) + F_{X,Y}(a, c).$$

  However, other probabilities generally require integration of the pdf.

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Example 3–4 (cont.)

Example 3–4 (cont.). Recall the pdf

$$f_{X,Y}(x,y) = \frac{x(1+3y^2)}{4} \quad \text{for } 0 \le x \le 2,\ 0 \le y \le 1.$$

Find the joint cdf and use it to compute $\mathsf{P}(X > 1, Y \le \frac{1}{2})$ again.

Solution: We can focus on the range of the random vector, namely the rectangle $[0,2] \times [0,1]$. Hence for $0 \le x \le 2$, $0 \le y \le 1$,

$$F_{X,Y}(x,y) = \mathsf{P}(X \le x, Y \le y) = \int_0^x \int_0^y \frac{u(1+3v^2)}{4}\,\mathrm{d}v\mathrm{d}u = \cdots = \frac{x^2(y+y^3)}{8}.$$

In this case, since X is less than 2 and Y is less than 1,

$$F_X(x) = F_{X,Y}(x,1) = \frac{x^2}{4} \quad \text{and} \quad F_Y(y) = F_{X,Y}(2,y) = \frac{y+y^3}{2}.$$

Check: These are proper cdfs with supports $[0,2]$ and $[0,1]$, respectively.

Next,

$$\begin{aligned}\mathsf{P}(X > 1, Y \le \tfrac{1}{2}) &= \mathsf{P}(1 < X < 2, 0 \le Y \le \tfrac{1}{2}) \\ &= F(2,\tfrac{1}{2}) - F(2,0) - F(1,\tfrac{1}{2}) + F(1,0) = \cdots = \frac{15}{64}.\end{aligned}$$

Compare with the earlier computation.

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Marginal CDFs and PDFs

- The *marginal cdf* for X is given by

$$F_X(x) = \mathsf{P}(X \leq x, Y < \infty) = \int_{-\infty}^{x} \int_{-\infty}^{\infty} f_{X,Y}(u, y)\, \mathrm{d}y \mathrm{d}u.$$

- Taking a derivative with respect to x, the *marginal pdf* for X is therefore

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, \mathrm{d}y, \quad \text{for all } x.$$

Think about the order of steps for finding marginal cdf's and pdf's.

- If you start with the joint cdf then typically you can next get the marginal cdf $F_X(x) = F_{X,Y}(x, \infty)$. Then take the derivative of that to get $f_X(x)$.
- If you start with the joint pdf then you usually will first find the marginal pdf $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, \mathrm{d}y$. And then you can integrate that to get $F_X(x)$.

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Example 3–4 (cont.)

Example 3–4 (cont.). Recall the joint pdf

$$f_{X,Y}(x,y) = \frac{x(1+3y^2)}{4} \quad \text{for } 0 \le x \le 2,\ 0 \le y \le 1.$$

Find the marginal pdfs and cdfs.

Solution: For $0 \le x \le 2$,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y = \frac{x}{4}\int_0^1 (1+3y^2)\,\mathrm{d}y = \frac{x}{4}(1+1) = \frac{x}{2}.$$

Therefore,

$$F_X(x) = \int_0^x \frac{u}{2}\,\mathrm{d}u = \frac{x^2}{4}, \quad 0 \le x \le 2.$$

Check: $F_X(x)$ is continuous and nondecreasing. Also, $F_X(0) = 0$ and $F_X(2) = 1$, which agrees with $\mathsf{P}(0 < X \le 2) = 1$.

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Example 3–4 (cont.)

Likewise, if $0 \le y \le 1$ then

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}x = \frac{(1+3y^2)}{4}\int_0^2 x\,\mathrm{d}x = \frac{(1+3y^2)}{2}$$

and

$$F_Y(y) = \int_0^y \frac{(1+3v^2)}{2}\,\mathrm{d}v = \frac{y+y^3}{2}.$$

Check: $F_Y(y)$ is increasing for $0 < y < 1$, and $F_Y(0) = 0$ and $F_Y(1) = 1$.

Reminder: when dealing with multiple random variables, distinguish their probability functions accordingly. In a given context, "$f(x)$" and "$f(y)$" are the same function!

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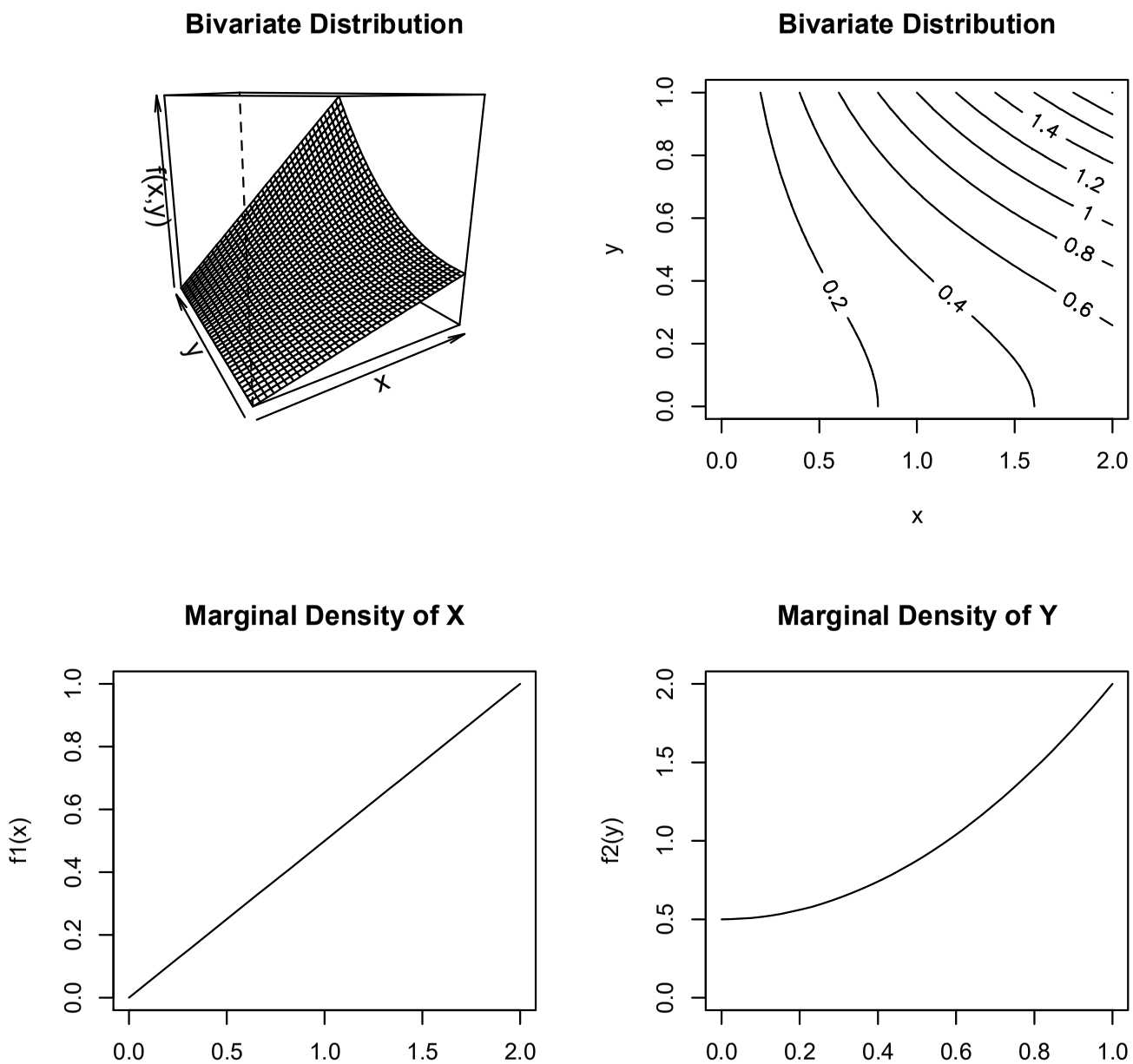
Example 3–4 (cont.)



Example 3–5. Suppose (X, Y) has joint pdf

$$f_{X,Y}(x,y) = \begin{cases} 3(x+y) & 0 \le x \le 1, 0 \le y \le 1, 0 \le x+y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Verify that this function is a pdf and find the marginal distributions.

Solution: We find (what should be) the marginal pdfs and then use them to show that the joint pdf is proper.

If $x < 0$ or $x > 1$ then $f_X(x) = 0$. Let $0 \le x \le 1$. Since $0 \le y \le 1 - x$,

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y = \int_0^{1-x} 3(x+y)\,\mathrm{d}y \\ &= 3[x(1-x) + (1-x)^2/2] = \frac{3(1-x^2)}{2}. \end{aligned}$$

Note that

$$\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = \int_0^1 \frac{3(1-x^2)}{2}\,\mathrm{d}x = 1.$$

This also shows that $f_{X,Y}(x,y)$ integrates to 1. (Why?)

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